

ECO 344 - Applied Econometrics I

Chapter 3

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- 2 Mechanics of Multiple Regression Model
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Multiple Regression Analysis

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- Now we can take out the ability from the error term and explicitly control for it.

But How do we measure ability? - measurement problem

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- What to do?

- Essentially the solution is to include all such variable in our model;

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$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \mu$$

- When we treat all other x as constant ($\partial x_k = 0$ for $k \neq j$), only then

$$\partial y = \beta_j \partial x_j$$

Mechanics of Multiple Regression Model

Derivation of OLS for Multiple Regression Model

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- we will use the same method to estimate our OLS estimators;

$$\text{Minimize : } \sum_{i=1}^n \hat{Error}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \hat{\beta}_3 x_{i3} - \dots - \hat{\beta}_k x_{ik})$$

$$\text{As } Error = y - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \hat{\beta}_2 x_2 - \hat{\beta}_3 x_3 - \dots - \hat{\beta}_k x_k$$

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- At the end we will get our beta estimates from solving these equations, only if we can uniquely find the solution of the above system.
- As long as there is no perfect collinearity among the regressors (M.L.R - III), we do find the unique solution of this system.

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- So if we want to see the effect of x_1 and x_2 on y then;

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Algebraic Properties of OLS in MLR

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- ③ The point $(\bar{y}, \bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_k)$ always lies on the regression line so ;

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- ④ The Sample covariance between each independent variable and residuals are zero. Similarly, the covariance between predicted values and residuals is zero.

$$\sum_{i=1}^n x_{ij} \hat{\mu}_i = 0 \quad \text{for each } j$$

Regression Through Origin

In case of multiple linear regression, when we force β_0 to be zero, we estimate the following model;

$$y = \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2 + \dots + \tilde{\beta}_k x_k$$

However,

$$E[\mu] \neq 0 \quad \text{Biased Estimators}$$

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- We can also use adjusted R-squared to address this issue. (Will discuss in later chapters)
- If we add a variable in the multiple regression model that has missing observation for 25 percent of the time. What would happen to R-Squared?

Comparing Simple & Multiple Regression

Another Perspective on Partial Effect

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$$wage = \delta_0 + \delta_1 \mu_2 + \mu_3$$

- This δ_1 will be the same as β_1 from the multiple regression model given below; - (**Frisch - Waugh Theorem**)

$$wage = \beta_0 + \beta_1 education + \beta_2 ability + \mu_4$$

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 - ② When $\gamma_1 = 0$ - There is no relationship between x_1 and x_2 .
- Back to $E[\mu|x] = E[\mu] = 0$ - There should not be any variable in error term that can affect y and also correlated with any x .

The Expected Value of OLS Estimators

Un-biasedness of OLS Estimators

Under Assumptions MLR-I to MLR-IV; $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \dots, \hat{\beta}_k$ will be unbiased estimators.

- 1 The population model is linear in parameters - M.L.R - I

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- 2 We have a random sample of n observations. - M.L.R - II

$$(x_{i1}, x_{i2}, \dots, x_{ij}, y) : i = 1, 2, 3, \dots, n$$

Un-biasedness of OLS Estimators

Under Assumptions MLR-I to MLR-IV; $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \dots, \hat{\beta}_k$ will be unbiased estimators.

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- Therefore, the regression coefficient will still be unbiased (Variance might be larger).

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- Let's look at MLR-I through MLR-IV ;

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$$E[\mu|x_1, x_2] \neq E[\mu|x_1, x_2, x_3] = 0$$

- Therefore, the regression coefficient will be Biased (Omitted Variable Bias).

- we know;

$$\hat{\eta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \gamma_1$$

Direction of Bias

- we know;

$$\hat{\eta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \gamma_1$$

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- Direction of Bias

TABLE 3.2 Summary of Bias in $\tilde{\beta}_1$ When x_2 Is Omitted in Estimating Equation (3.40)

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
$\beta_2 > 0$	Positive bias	Negative bias
$\beta_2 < 0$	Negative bias	Positive bias

The Variance of OLS Estimators

Homoscedasticity

- The error μ has the same variance given any value/combination of the explanatory values. - M.L.R - V

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- OR we can also use some statistical test to determine homoscedasticity/heteroscedasticity. (Will be covered in later chapters).

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- In STATA, we visualize the error with independent variable to see the pattern in variance.
- OR we can also use some statistical test to determine homoscedasticity/heteroscedasticity. (Will be covered in later chapters).
- The assumptions M.L.R - I through M.L.R - V are collectively known as **Gauss Markov Assumptions**.

Variance of the OLS Slope Estimators

- Under Assumptions M.L.R - I through M.L.R - V, the variance of the slope estimators would be;

$$Var[\hat{\beta}_j] = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

where σ^2 is the variance of error.

$$SST_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

R_j^2 is the R-Squared from the regression of x_j on all the remaining independent variables.

The Variance of Error Term

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- Therefore, there are three components of the variance of β_j estimators.

The Variance of Error Term

1. The Variance of Error Term

- We can estimate the variance of the residual term using the following equation;

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n \hat{\mu}^2}{(n - k - 1)} = \frac{SSR}{(n - k - 1)}$$

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- The degree of freedom is the degree to which your observations are more than the technically required values.
- As the $\hat{\mu}$ is from the sample the $\hat{\sigma}^2$ is also an estimator. It will be unbiased under the assumption M.L.R - I through M.L.R-V.

$$E[\hat{\sigma}^2] = \sigma^2$$

The Total Sample Variation in x_j

2. The Total Sample Variation in x_j



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- The bigger the variance is, the less variance of the slope estimators.
(Ideal)

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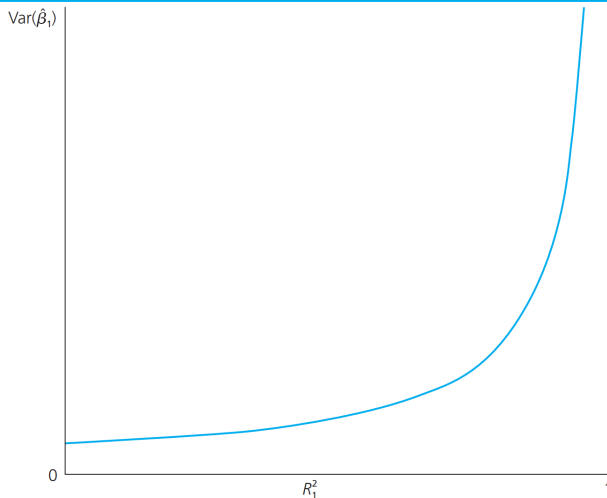
- The bigger the R_j^2 is, the bigger variance of the slope estimators.

The smaller the R_j^2 is, the smaller variance of the slope estimators.
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The Linear Relationship among Independent Variables

3. R_j^2 The Linear Relationship among Independent Variables

FIGURE 3.1 $\text{Var}(\hat{\beta}_1)$ as a function of R_1^2 .



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- Exact relationship between male and female;

$$Male + Female = 1$$

In this case the regression would not run.

Variance Inflation Factor - VIF

- Let's Define Variance Inflation Factor $\frac{1}{1-R_j^2}$ is the component of variance that depends on the covariance between independent variable.

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- Ideally, we want to have the VIF to be as low as possible (Low R_j^2).
- However, it is not in our hand when we are trying to remove endogeneity.
- One trick is to only care about the our main variable while looking at the VIF.

Variances in Misspecified Models

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- And our misspecified model would be;

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- Therefore, there is a clear trade off between Bias and Variance.
- Generally, we care more about variance in forecasting and care more about bias while finding the effect of x on y.

Gauss Markov Theorem

Gauss Markov Assumptions

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- When the standard set of assumptions holds, we need not look for alternative unbiased estimators (Minimizing other forms of errors): none will be better than OLS.

Examples

Example 1:

- 1 Using the data in GPA2 on 4,137 college students, the following equation was estimated by OLS:

$$\widehat{colgpa} = 1.392 - .0135 \text{ hspc} + .00148 \text{ sat}$$
$$n = 4,137, R^2 = .273,$$

where *colgpa* is measured on a four-point scale, *hsperc* is the percentile in the high school graduating class (defined so that, for example, *hsperc* = 5 means the *top* 5% of the class), and *sat* is the combined math and verbal scores on the student achievement test.

- (i) Why does it make sense for the coefficient on *hsperc* to be negative?
- (ii) What is the predicted college GPA when *hsperc* = 20 and *sat* = 1,050?
- (iii) Suppose that two high school graduates, A and B, graduated in the same percentile from high school, but Student A's SAT score was 140 points higher (about one standard deviation in the sample). What is the predicted difference in college GPA for these two students? Is the difference large?
- (iv) Holding *hsperc* fixed, what difference in SAT scores leads to a predicted *colgpa* difference of .50, or one-half of a grade point? Comment on your answer.

Example 2:

- 7 Which of the following can cause OLS estimators to be biased?
- (i) Heteroskedasticity.
 - (ii) Omitting an important variable.
 - (iii) A sample correlation coefficient of .95 between two independent variables both included in the model.

Example 3:

- 15 The following estimated equations use the data in MLB1, which contains information on major league baseball salaries. The dependent variable, *lsalary*, is the log of salary. The two explanatory variables are years in the major leagues (*years*) and runs batted in per year (*rbisyr*):

$$\widehat{lsalary} = 12.373 + .1770 \text{ years} \\ (.098) \quad (.0132)$$

$$n = 353, \text{ SSR} = 326.196, \text{ SER} = .964, R^2 = .337$$

$$\widehat{lsalary} = 11.861 + .0904 \text{ years} + .0302 \text{ rbisyr} \\ (.084) \quad (.0118) \quad (.0020)$$

$$n = 353, \text{ SSR} = 198.475, \text{ SER} = .753, R^2 = .597$$

- (i) How many degrees of freedom are in each regression? Why is the SER smaller in the second regression than the first?
- (ii) The sample correlation coefficient between *years* and *rbisyr* is about 0.487. Does this make sense? What is the variance inflation factor (there is only one) for the slope coefficients in the multiple regression? Would you say there is little, moderate, or strong collinearity between *years* and *rbisyr*?
- (iii) How come the standard error for the coefficient on *years* in the multiple regression is lower than its counterpart in the simple regression?